



Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU

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LANDSLIDE AND FLOOD THREAT ADVISORIES

To: **Brgy. Chairperson
Brgy. Barangka
City of Marikina
Metropolitan Manila**

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Riverbanks Mall (Old Barangay Hall)	14°37'50.1" N 121°04' 44.9" E	High	Brgy. Barangka is generally situated on sloping terrain and most of its areas experiences low level flooding caused by rainfall runoff. However, those situated on relatively flat plains are subjected to worst flood levels.
Health Center	14°38' 06.1" N 121°04' 53.2" E	High	The overflow of the Marikina River is the primary source of flooding. The said channel is notorious to overflow during inclement weather conditions and floodwaters reach Brgy. Barangka. Flood heights generally varies from area to area but worst flood heights recorded were during Typhoon Ondoy with more than ~2m floods. This is particular in an area called 'Urban Bliss Barangka.' Further, more than ~1 m flood heights were reported in the Riverbank Malls.
Urban Bliss Barangka	14°37' 50.1" N 121°04' 44.9" E	Very High	
	14°37' 51.7" N 121°04' 44.8" E	Moderate	
	14°37' 53.4" N 121°04' 44.6" E	Moderate	

Remarks / Recommendations:

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.

- ✓ **Construction and/or improvement of drainage system.** Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Landslide Susceptibility Rating	Geohazards Assessment
Dela Costa Homes	N 14°38'13.3" E 121°04' 50.4"	High	There are isolated steep slopes/ridges in Brgy. Barangka that was probably created from human intervention primarily for land design (i.e. cut slope). These areas are susceptible to mass wasting, particularly rock fall, due to their sheer steepness. Such situation was observed in a parking lot in Dela Costa Homes. Although no major landslide (rock fall) incident had been reported, the CDRRMO noted instances of pebble to cobble sized rocks chipping off the exposed slope. The said slope is an exposure of bedded adobe with visible jointing. Some fractures are noted to be widening (30 cm) with a visible talus fan. The area is currently being used as a parking lot for residents. Adjacent to the lot are stairs leading to
Sto. Niño Park	N 14°38'20.0" E 121°04' 49.8"	High	

			the upper portion of the said slope; here, some areas exhibit differential weathering leading to overhanging of some of the bedded adobe. These are areas which may fail in the future. Some structural mitigation is recommended.
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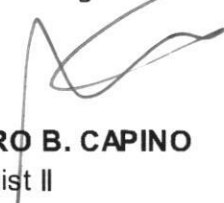
Remarks / Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
- ✓ **Observe for saturated ground and/or displaced roads surfaces** and report to the MGB/local authorities.
- ✓ **Implementation and maintenance of appropriate slope stabilization** (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended.
- ✓ **Barangay Disaster Risk Reduction Management Team must implement monitoring and pre-emptive evacuation** during monsoon and typhoon season where extreme heavy rainfall occurs.
- ✓ **Activate** Barangay Disaster Risk Reduction Management Team in cooperation with the City Disaster Risk Reduction Management Team during monsoon and typhoon season.
- ✓ The CDRRMO together with the BDRRMO should conduct information and education awareness on this particular type of geohazard, particularly on landslide hazard to the landslide-prone communities.

Prepared by:


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MA. LOURDES STEPHANIE V. LEIDO
Senior Geologist

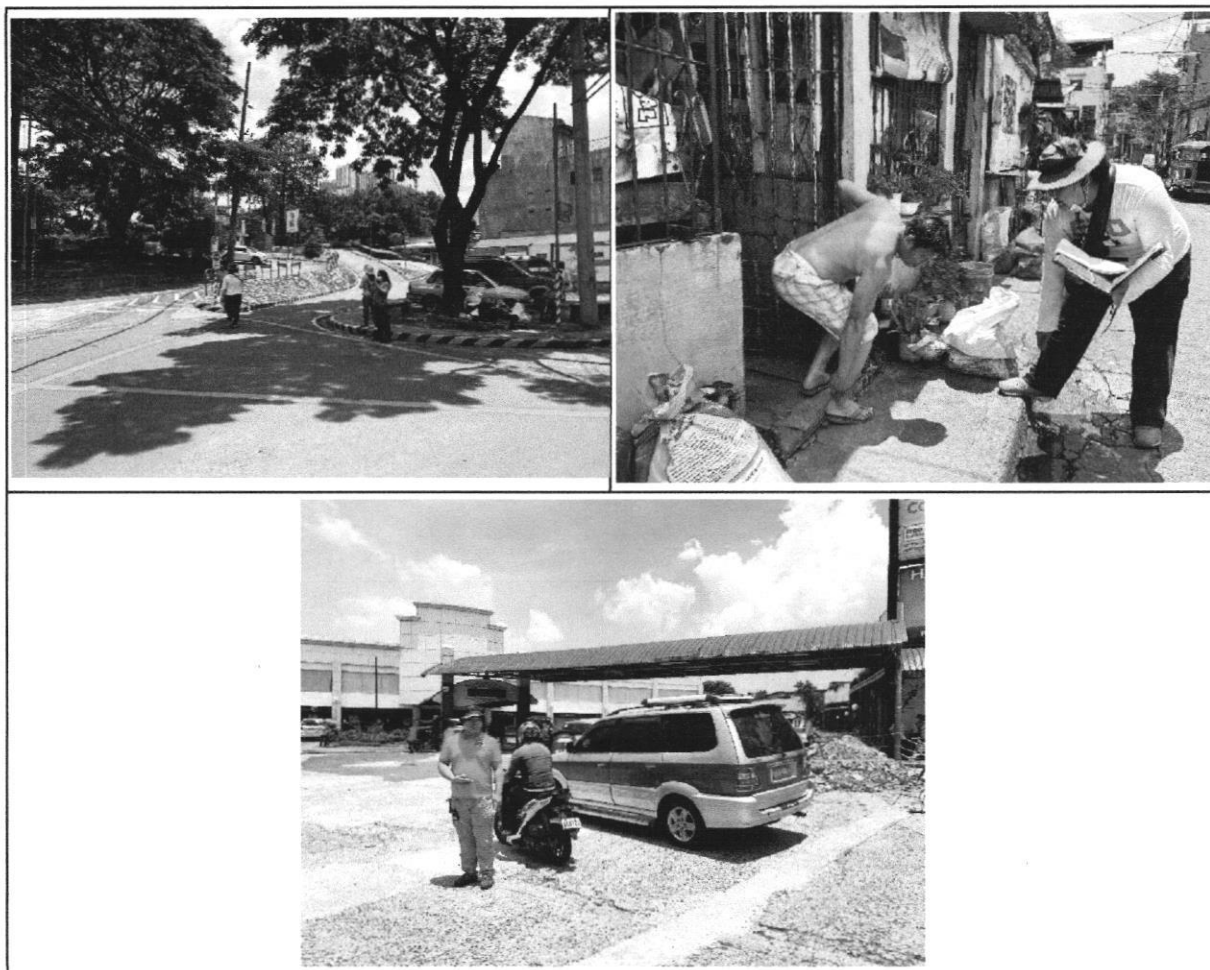

JETHRO B. CAPINO
Geologist II

Received by:

Designation:

Office:

PHOTOS



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Recommendations:

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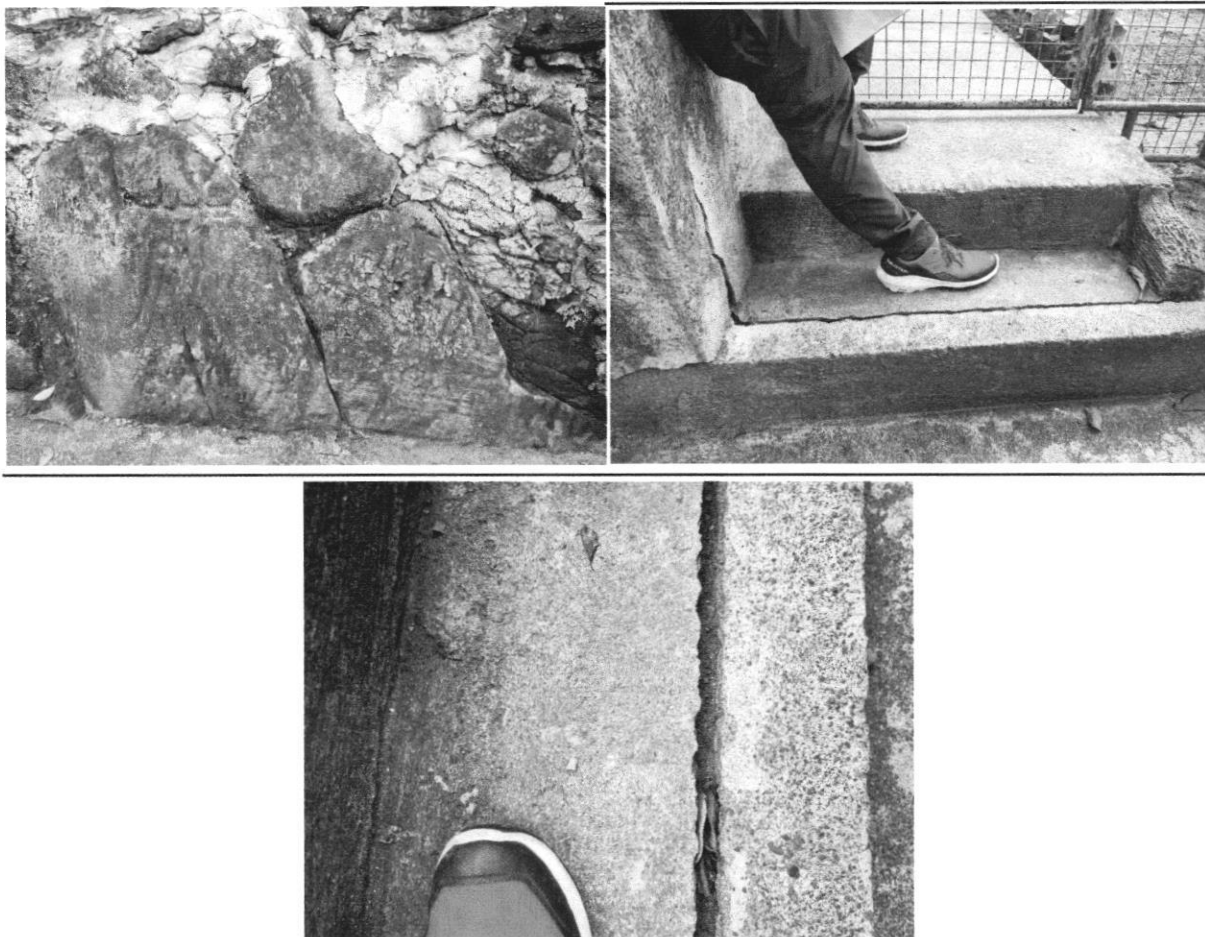




Such situation was observed in a parking lot in Dela Costa Homes. Although no major landslide (rock fall) incident had been reported, the CDRRMO noted instances of pebble to cobble sized rocks chipping off the exposed slope. The said slope is an exposure of bedded adobe with visible jointing. Some fractures are noted to be widening (30 cm) with a visible talus fan. The area is currently being used as a parking lot for residents. It is recommended to put a warning signage to warn car owners that the area is prone to rock fall. (GPS coordinates: N 14°38'13.3", E 121°04' 50.4")

Recommendations:

- ✓ **Observe presence of mass movement** (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB.
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Adjacent to the lot are stairs leading to the upper portion of the said slope; here, some areas exhibit differential weathering leading to overhanging of some of the bedded adobe. These are areas which may fail in the future. Some cracks in the pavement was also observed. (GPS coordinates: N 14°38'20.0", E 121°04' 49.8")

Recommendations:

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